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Field Testing and Study of Horizontal Water Injectors in Increasing Ultimate Recovery from a Reservoir in Thamama Formation in a Peripheral Water Injection Scheme in a Giant Carbonate Reservoir, United Arab Emirates.

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1. Abstract.

Development trials of peripheral horizontal injectors were performed in an attempt to establish waterflood sweep within the low permeability lower half of a Thamama reservoir. This paper describes the performance of the horizontal injectors, as well as a time lapse formation tester interference test between a horizontal injector and open hole observation well, spaced 100 meter apart. The test was designed to measure the distance it takes water from the horizontal injector to break through into the high permeability upper half of the reservoir.

Interpretation of the results show that water from horizontal injectors moves vertically out of zone within 100 meters of the injection point, and is not contained within the lower half of the reservoir. Full field simulation after history matching the interference test results shows small increase in vertical and areal sweep efficiency resulting from utilization of horizontal injectors. Radioactive tracer injected into a horizontal injector in the lower part of the reservoir migrated upward into the upper part and traveled up dip 1km to an observation well within 3 years, traveling more than 3 feet per day.

While use of horizontal producers has been identified as the primary method for increasing recovery from the lower part of this Thamama reservoir, the benefits of horizontal injectors over vertical injectors is small. Horizontal injectors can only be justified when costs of adding a horizontal tail to a newly

drilled injector is equally small. High recompletion costs of converting existing peripheral vertical injectors to horizontal injectors through sidetracking is not justifiable, or economic.

2. Introduction. The field is located in the onshore area of Abu Dhabi about 110 km to the Southeast of Abu Dhabi Island^{1,2} and was discovered in 1965 in the Thamama Reservoirs. The field is an elongated faulted, doubly plunging anticline with the longer axis trending Northeast - Southwest. It is about 30 km long and 10 km wide. At top Thamama level, the flanks dip 3-5° on the western and eastern side and 1 - 2° on the northern and southern side.

This Thamama reservoir (Upper Kharaib Fm.) is the main oil bearing and producing reservoir in the field with light 40 API sweet crude. It can be subdivided roughly into an "upper" and a "lower" unit based on their depositional facies and reservoir properties, predominantly change of permeability (Fig. 1), with oil in place divided evenly between the two. The reservoir has been further subdivided into five units based upon the presence of dense stylolitic intervals that separate the reservoir rock (Fig 1). The dense units do not act as barriers to fluid flow or to pressure, but thicken from crest to flank of the structure. The thickness of the reservoir ranges between 191 ft in the crestal area and 165 ft down flank, with average porosity and arithmetically averaged permeability vary from a maximum of 34% and 363 mD in the crestal area to a minimum of 20% and 10 mD down flank.

Wettability of the reservoir is thought to become progressively more oil-wet from flank to crest, with initial water saturation ranging from 5 s.u. to 10 s.u. at the crest, depending upon rock quality, with increasing saturation toward the flank through a significant transition zone.

3. Production and waterflood history. The field was brought on line in 1973, with peripheral water injection starting in 1976. Initially all producers and injectors were commingled across the reservoir, and shortly after the

initiation of the waterflood it was found that almost all the injection was taken by the high permeability upper unit of the reservoir.

A full field re-completion program was implemented in 1982 to convert all producers and injectors to dual completions to produce the upper and the lower units separately while maintaining the peripheral injection scheme. In conjunction with the dual completions, injection into the highest permeability subzone in the upper unit was restricted along the East and West flanks in order to retard the advance of water toward the crest of the field. This restriction still remains in place.

In 1995 a campaign to convert watered out vertical wells to horizontal producers from the lower unit was initiated (Figure 1). To this point all wells had been vertical. The goal of the program was to re-establish production from the lower unit lost due to wells which had died due to water coning from the upper unit down into the lower unit and killing the well⁵. Wells typically died below 50% water cut due to the absence of artificial lift in the field. Horizontalization of vertical producers continues to this day, along with infill drilling of new horizontal wells, as the main strategy for increasing recovery from the lower unit, with over 60 producers of this type in place.

Monitoring data from 1985 to present has shown that water injected selectively into the lower unit completions along the periphery of the field has broken out of zone, upward into the high permeability upper unit, without any sweep benefits in the lower unit. Hence, in 1997 horizontal injectors were proposed as a possible tool for improving sweep in lower unit. Potentially, the sweep increase would be accomplished by distributing the injection over a larger area, minimizing the pressure differential between the upper and lower units at the injector, resulting in the injection water staying longer in lower unit. Six horizontal injectors were added in 1998 along with a monitoring program to determine the benefits of these wells to production.

4. Mechanisms controlling production and injection performance. The main characteristics of the production history, and the mechanisms thought to influence these characteristics are as follows:

- a) Water thieving and holdup in the high permeability upper unit due to capillary pressure effects.
- b) Inverse or reverse conning, a rapid process where drawdown from a lower unit producer overcomes capillary pressure and relative permeability effects which accelerate the downward movement of water from top to bottom of the reservoir.
- c) Water slumping due to gravity, a slow process evident over 30 years of production history.

Figure 1 visually depicts the main production characteristics of this field over the thirty-year field life. With dual injectors water injected into the lower unit completion migrates rapidly into the upper unit. No evidence of injected water in the lower unit has ever been found, even though 90% of injection for the past 15 years have been through the lower completions. The injection water is held up in the highest permeability subzone through means of oil wet capillary pressure behavior similar to that described in other local reservoirs^{3,4}. As the water advances toward the crest of the structure, the vertical wells water out, first the upper completions, and shortly after the lower completions as the water is sucked down, typically within months of the upper completion encountering water. Again, with no artificial lift each completion typically stops producing near 50% water cut.

Each non-producing vertical well is then sidetracked, creating a horizontal section generally 2500 to 3000 feet long, targeted 25 feet above the base of the reservoir. The well is sidetracked over 700 feet away from the original vertical completion in order to avoid intersecting the water cone that extends into the lower unit (Figure 2). The shape and extent of cones have been determined through investigations using multiple sidetracks at various distances from vertical wells, and from use of observation wells located to monitor long term horizontal producer performance. Once a horizontal sidetrack is put on production, water begins coning toward the base of the reservoir between 1 to 2 feet per month, depending upon the distance from the horizontal hole (Figure 3). Dry oil is produced until approximately 2 mmstbo cumulative recovery is reached. The magnitude of the capillary pressure holdup is small, less than 10 psi; therefore, the smallest drawdown from a producer will initiate the water coning.

Slow downward movement of water due to gravity slumping is also a critical mechanism operating throughout the reservoir, identified by saturation logs acquired where there is no drawdown associated with producers. Numerous open hole logs have been acquired from wells targeting deeper reservoirs that show a wedge shape to the held-up water, with the thickness of the wedge increasing toward the flank (Figures 1 and 4). Toward the flank the water has slumped into the top of the lower unit. It is clear from abundant surveillance data that with enough time gravity will also overcome capillary pressure effects and pull water down toward the base of the reservoir. Slumping is apparent everywhere but is more noticeable toward the flank where the reservoir is more water wet, and where injection water has resided for the longest period of time. Gravity slumping may not begin until Sor in the high permeability layers is sufficiently reduced through sweep to cause the hydrostatic weight of the influx water column to exceed a threshold capillary pressure, after which the downward water

movement begins. Therefore, gravity slumping would be fastest toward the field flanks where residuals are lowest due longer exposure and proximity to the peripheral injectors. However, the gravity slumping process is slow, and over 25 years the entire lower unit remains essentially injection water free.

5. Performance of vertical and horizontal injectors

The average injectivity index for the six horizontal injectors drilled in 1998 showed a four-fold increase over the original vertical injectors that were replaced (Injectivity index of 6.8 psi-bbl vs. 1.5 psi-bbl). The injection performance of the horizontal wells has been steady with no significant loss in injectivity with time yet noted. Injection profiles were run on five of the injectors, two that also had tubing conveyed image logs. Figure 5 shows the data from one well that shows the presence of open or closed fractures along the horizontal wellbore. There is no evidence that the fractures influence the injection profile or that thiefing is present. Typically in the horizontal injectors the first several hundred feet of the hole have lower injectivity, where due to lack of robust depth control during drilling the well intersected the very base of the reservoir where permeability is lowest.

6. Time-lapse repeat tester interference test.

Goal. One of the goals of horizontal water injectors was to reduce the upward vertical movement of the water into the high permeability upper unit, thereby increasing the vertical and areal sweep in the lower unit. An interference test was designed in order to determine if this goal was achieved, and the test implemented in 1999.

Interference Test Design. A vertical injector was converted to horizontal in December 1998 and brought on injection by February 1999. Radioactive Tritium was added to the injection stream at this time. The deviated sidetrack was completed with 7" casing across the upper unit to 25 feet above the base of the lower unit, where the bare-foot horizontal section begins. Prior to initiation of full rate injection the quality of the cement behind the 7" casing was verified using pulsed-neutron-logging techniques to ensure that no channeling was responsible for the upward movement of water. Logging confirmed excellent cement quality, and no channeling was observed at 10,000 bwpd water injection rate.

The interference test was performed in October 1999, after 8 months of continuous injection, utilizing a vertical observation well that was drilled 100 meters offset from horizontal injector #1 (Figure 6). Observation well #1 was drilled through the reservoir and left open while a series of time-lapse pressure measurements were made using the Reservoir Characterization Instrument (RCI). All surrounding injectors, including the horizontal, were shut in several days before the first pressure measurements were

made. Approximately 800 psi overbalance was present during the testing phase across reservoir. A full suite of open hole logs, including image logs were run on the observer to complement the logs run on horizontal injector #1, and to allow for correlation, and quantification of the thickness, of the dense stylolitic intervals between the two wells.

Three sets of pressure measurement across reservoir were recorded, two baselines and one after initiation of injection into the horizontal injector (Figure 7). Pressure points were repeated twice at each station and confirmed to within 0.1 psi of the first measurement. Approximately 1 hour was required to complete each set of pressure points.

Interference Test Results. The first baseline set of pressures fell on a straight-line gradient of 0.51 psi/ft, reflecting the location of the well in nearly 100% water where formation and injected water are have approximately the same salinity, over 150,000 ppm Cl. Figure 8 shows the three series of pressure measurements, as well as the location and thickness of the dense stylolitic intervals that were interpreted from image logs (thin grey/red lines).

The second set of baseline pressures, taken 12 hours after the first, fall on a equivalent gradient, and show the reservoir pressure decreasing at 0.2 psi/hr equally across all the reservoir subzones. Injection in the horizontal injector was initiated at 800 bwpd immediately after the second baseline set of points was taken.

The final set of pressure points was taken after 96 hours of injection. The stratigraphic location of the horizontal injector, located 100 meters away, is indicated at the lower right hand side of Figure 7. Clearly, large pressure increases are noted across the entire reservoir interval, from top to bottom. The dynamic pressure gradient shows no indication of barriers to vertical fluid movements, and a minor baffling effect is present across the DB2 dense body.

7. Simulation of interference test. A 250k black oil model with local grid refinement was used to history match the results of the interference test, and then to extrapolate the results in prediction mode to estimate the potential benefits of the horizontal injectors on ultimate recovery. The full field model used was matched for the 30 years of development history with 300 x 300 meter cells which produced a water saturation match shown in track A of Figure 8. Saturation functions used to simulate the water hold up are similar to those described in previously reported work^{3,4}. Upscaled permeability and porosity at the observation well location were compared to logged values, and to core data from offset wells Observer #2 and Injector #4 (Figures 6 and 8). Minor adjustments were made to the dense body permeability in the local area to bring the upscaled values in line with the core data (Track B, Figure 8).

Local grid refinement was added to the full field model in order to properly locate the wells in the model, which are only 100 meters apart. Twenty-five meter cell size was used, leaving three blocks separating the observer and injector.

Track C of Figure 8 shows the match of the interference test over the 5-day period. A good match is apparent across all of the reservoir for all three sets of pressures. Some small differences are noted at the oil water contact, where the history matched and actual water saturation are different by 10%. The pressure differences may therefore be due to differences in the total compressibility across this interval.

8. Predicted Horizontal Injector Performance. Based upon the matched interference test from the horizontal injector, the model was run 30 years into the future to determine the benefits on ultimate recovery. Tracer was added to the model to better visualize the movement of injected water, and to compare with the movement of the actual radioactive tracer added to the horizontal injector.

The location of the horizontal injector, the open hole observer, and a cased hole observer 1 km up dip are displayed in Figure 6. Production in the model comes from a series of vertical and horizontal wells located up-dip of the LGR, part of the global grid of the full field model not shown here.

The model shows that the injection water breaks through into the upper unit within 100 meters of the injector. Once the breakthrough occurs, the water races up-dip along the high permeability upper unit with minimal containment in the lower unit. After 30 years of continuous injection, the traced water has moved less than 500 meters, with the majority of movement below the original oil water contact in the lower unit.

Tracer was predicted to have reached the second observer within 3 years. In fact, the radioactive tritium was recovered from the upper unit in June, 2001, showing a movement of greater than 3 feet per day for the water injected into the lower unit to migrate upward into the upper unit and travel 1 km to the observer.

9. Horizontal vs. Vertical Injector Comparison. Injection using a vertical injector perforated at the base of the lower unit was simulated as a baseline for comparing the horizontal injection results. There is a very small difference between the two cases, even though the volume of injection from the horizontal injector was nearly twice that of the vertical. Water in the horizontal case advanced approximately 100 meters more than the vertical case, and there was more difference in the length of the flood front. However, the location of both injectors below the original OWC means that these small differences in the flood

geometry result in even smaller sweep benefits. Cases were also run with injection into the horizontal injector restricted to 4000 bwpd with no appreciable change over the high rate case. In addition, cases were run after adding a horizontal producer within the up-dip section of the LGR, again with the same results as the first prediction cases.

In total, the horizontal injector cases resulted in approximately 100,000 bbls of additional recovery over a 30-year production period.

10. Cost and economic evaluation of injector benefits. The incremental cost of working over existing vertical injectors and converting them to horizontal injectors in the lower unit is \$2.0 mm. Incremental costs of adding a horizontal tail to a newly drilled injector are only \$0.5 mm. The simplest comparison on a dollar per barrel basis shows that converting existing peripheral vertical injectors into horizontals through sidetracking is not economically viable (\$20/bbl). However, if the intention is to drill a new peripheral injector, adding a horizontal tail to the well appears moderately beneficial, at \$5.0/bbl.

11. Conclusions. Simulation of a time lapse interference test shows that water injected into the lower unit of the reservoir in either vertical or horizontal water injectors along the periphery of the field breaks into the upper unit thief zone and adds little benefit to sweep of the lower unit. Radioactive tracer injected into a horizontal injector in the lower unit broke into the upper unit and traveled up dip 1km to an observation well within 3 years traveling at more than 3 feet per day. Conversion of existing peripheral vertical water injectors into horizontal injectors through sidetracking is not economically viable given the high cost and small incremental production benefits over the long remaining field life (over 30 years).

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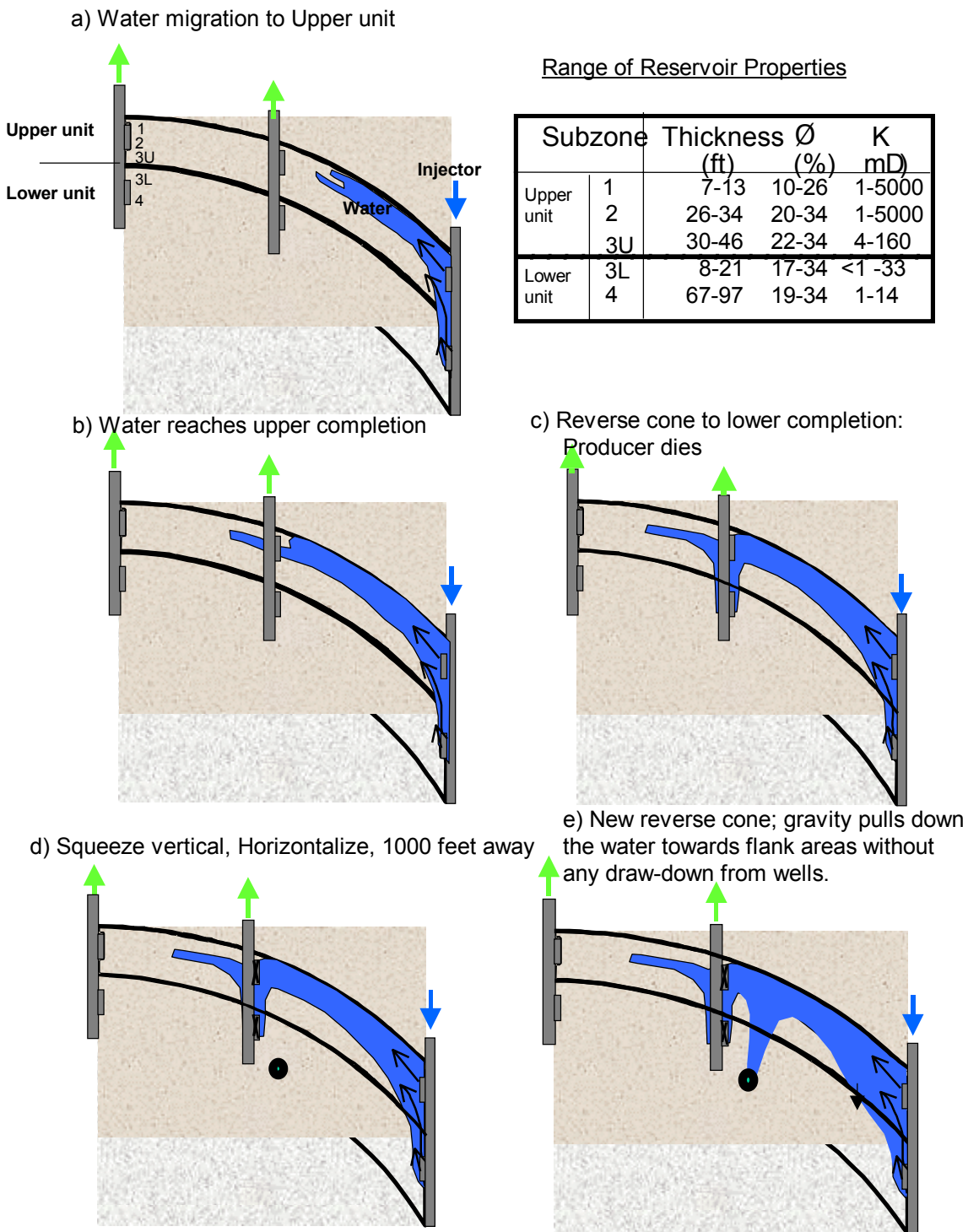


Figure 1: Production and injection history, and reservoir

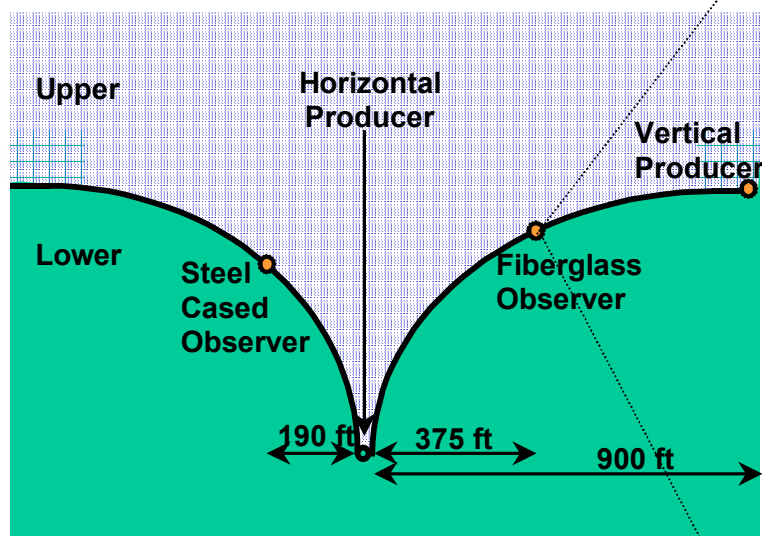


Figure 2: Shape of inverse cone from horizontal well.

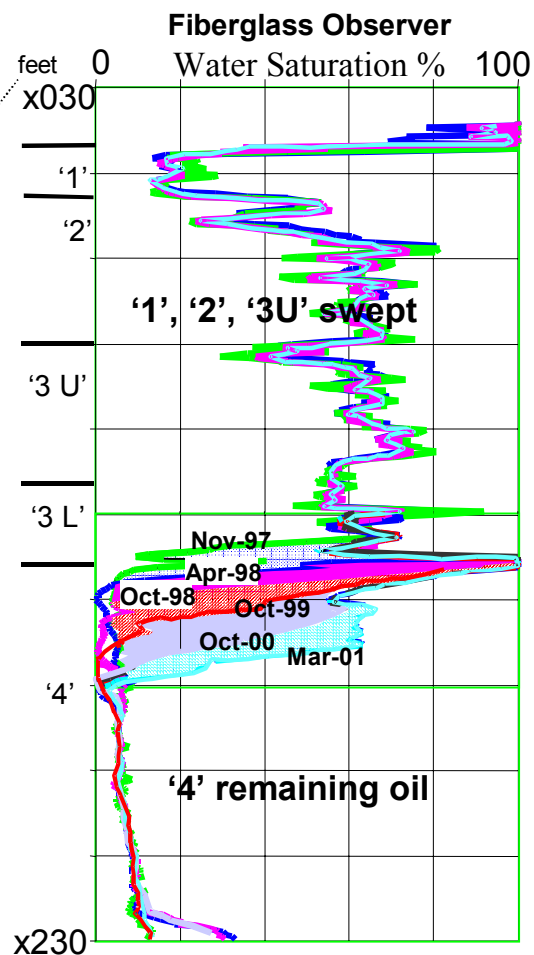


Figure 3: Rate of coning induced by horizontal producer.

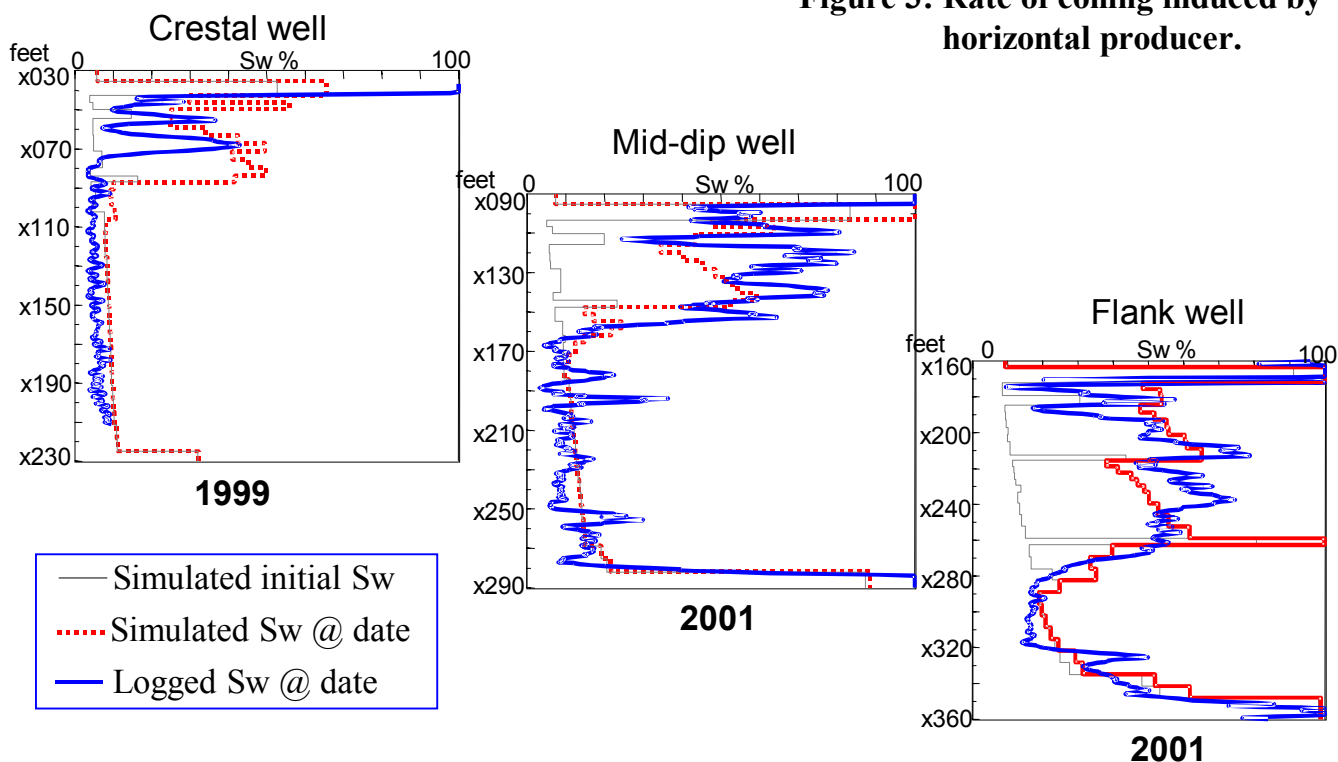


Figure 4: Wedge geometry from crest to flank induced by gravity on water.

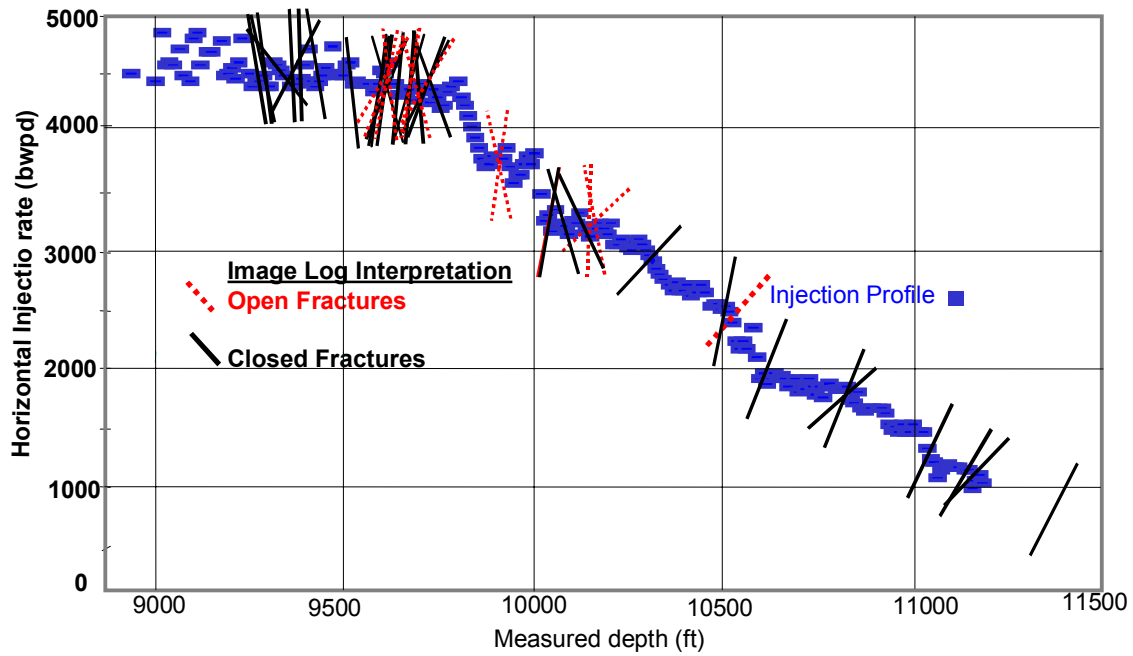


Figure 5: Injection and image log of a horizontal injector.

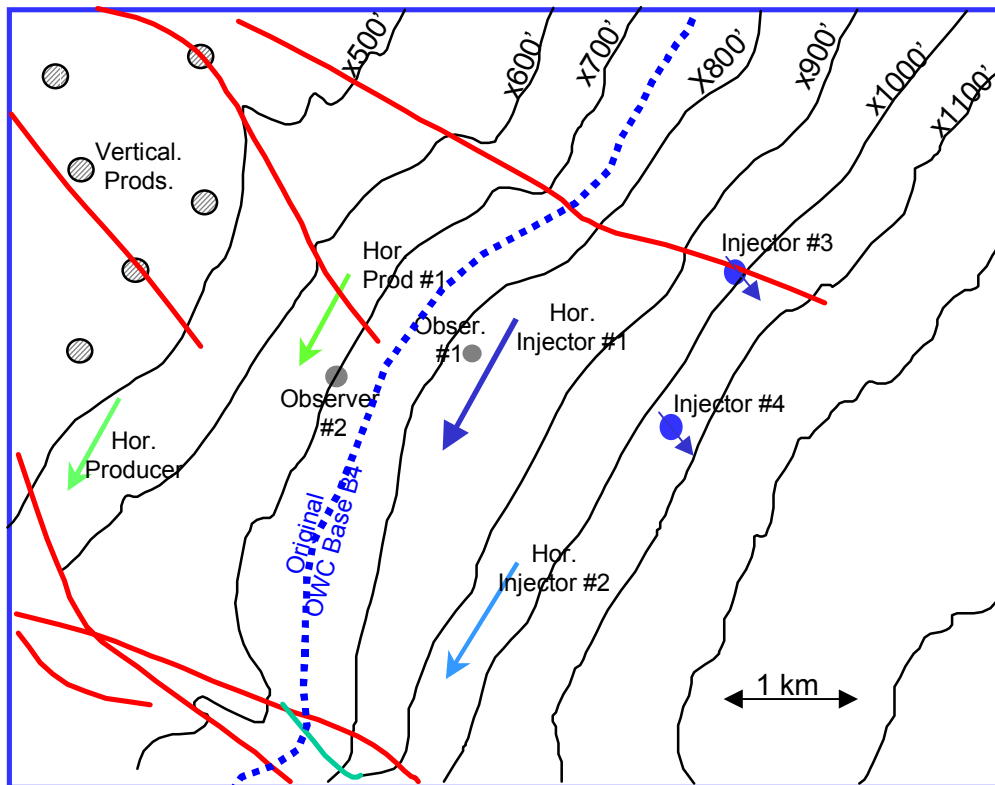


Figure 6: Location map of interference test.

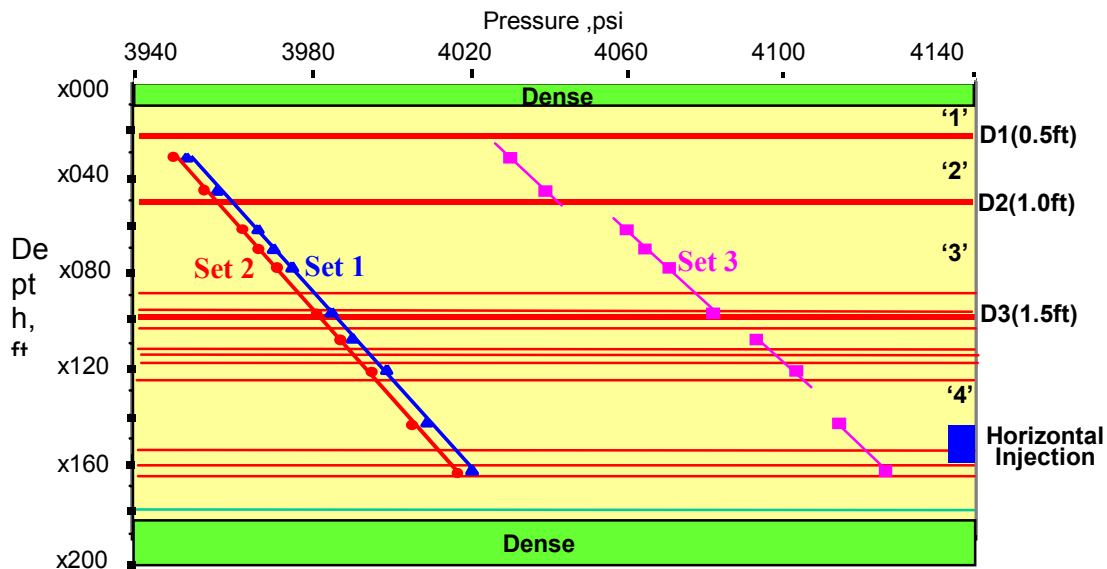


Figure 7: Pressure points taken during interference test.

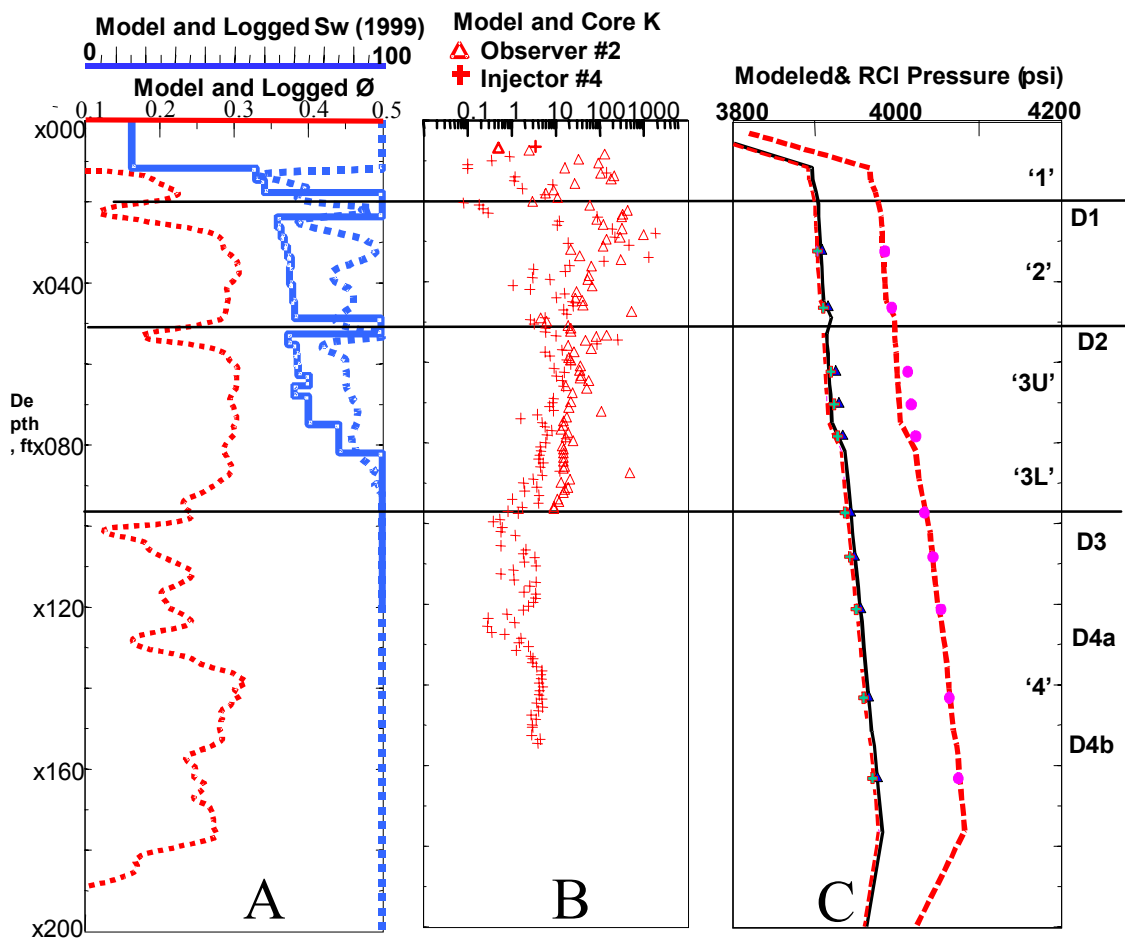


Figure 8: History match results of interference test.